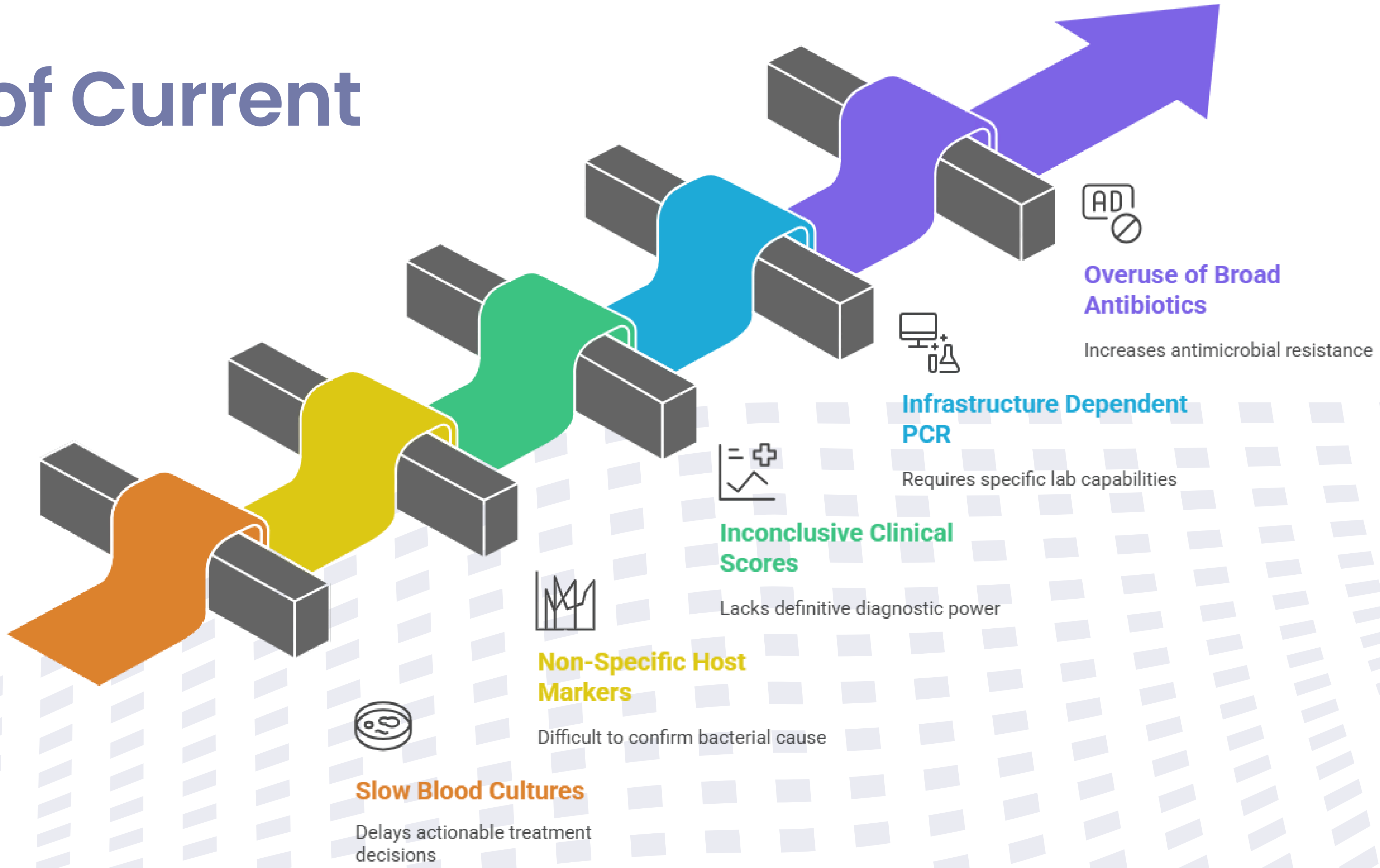
SERS-Based Nanobiosensor Workflow for Sepsis Diagnostics

Clinical Urgency of Sepsis

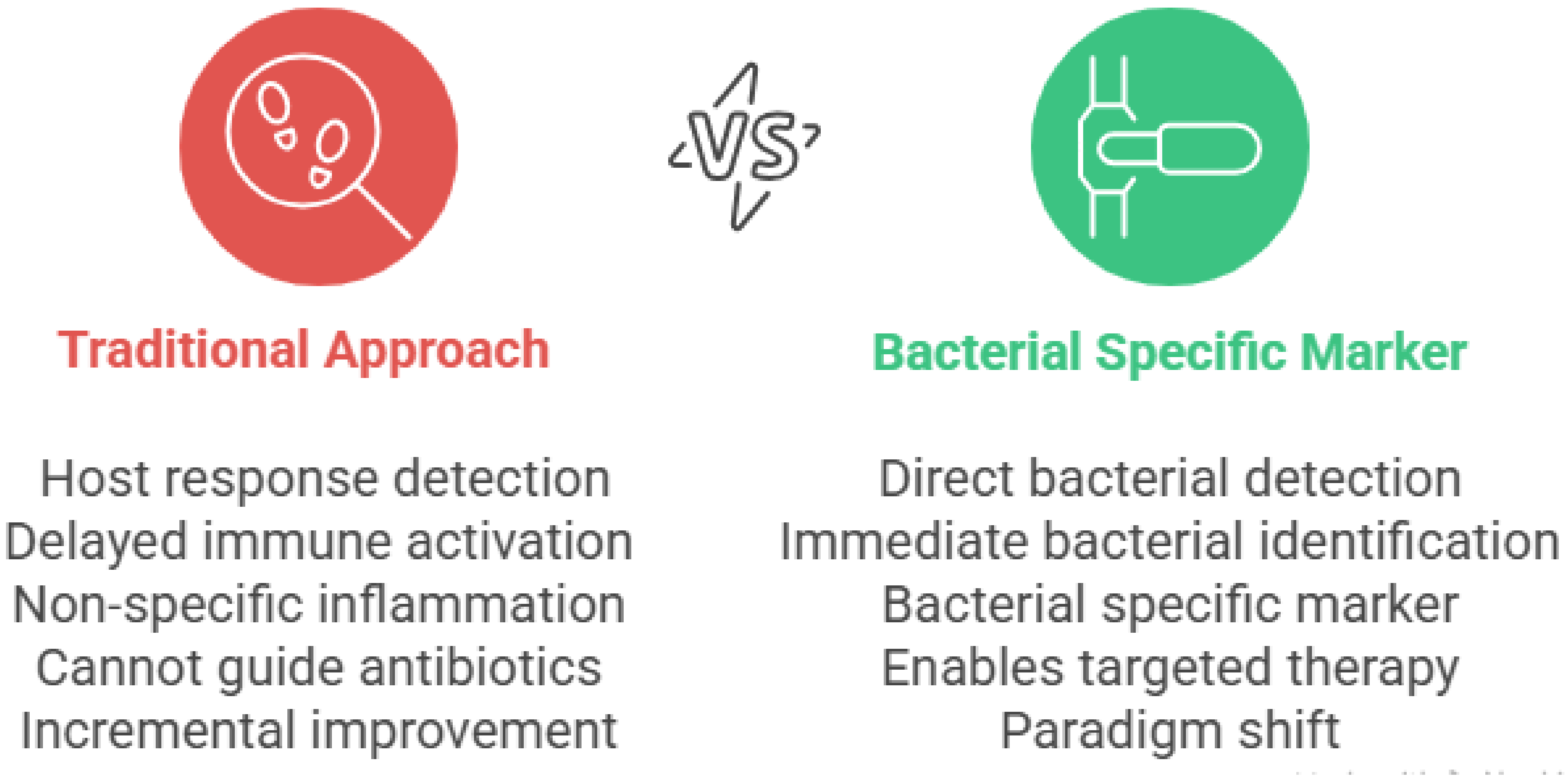
- Approximately 49 million sepsis cases globally (2020) with ~11 million deaths.
- Early detection and treatment are critical; delays increase mortality.
- Sepsis is a major contributor to hospital fatalities and ICU burden.

Limitations of Current Diagnostics



Why Bacterial-Specific Biomarkers?

- Early diagnosis and improved clinical outcomes.
- Directly indicate the presence of bacteria (not just host inflammation).
- Enable Gram classification. Reduce unnecessary broad-spectrum antibiotic use and combat AMR.



Difference	Gram-Positive	Gram-Negative	Clinical Significance	Reference
Cell Wall Structure	Thick peptidoglycan (20-80 nm)	Thin peptidoglycan + outer membrane	Determines antibiotic penetration and SERS targeting strategy	Silhavy et al. Cold Spring Harb Perspect Biol 2010;2(5):a000414
Peptidoglycan Content	90% of cell wall mass	5-10% of cell wall mass	Major structural target for β-lactam antibiotics and detection	Vollmer et al. FEMS Microbiol Rev 2008;32(2):149-167
Outer Membrane	Absent	Present (barrier function)	Primary resistance mechanism; barrier to drugs and immune system	Nikaido H. Microbiol Mol Biol Rev 2003;67(4):593-656
Lipopolysaccharide (LPS)	Absent	Present (endotoxin component)	Major sepsis mediator; triggers cytokine storm and shock	Raetz & Whitfield. Annu Rev Biochem 2002;71:635-700
Gram Staining	Purple/Violet (retains stain)	Pink/Red (loses stain)	Primary laboratory identification method (95% accuracy)	Gram HC. Fortschr Med 1884;2:185-189
Antibiotic Susceptibility	Generally, more susceptible	More resistant (outer membrane)	Guides empirical antibiotic selection in sepsis management	Delcour AH. Biochim Biophys Acta 2009;1794(5):808-816
Endotoxin Production	No endotoxin	Lipid A endotoxin present	Key pathophysiology difference in sepsis severity and outcomes	Rietschel et al. FASEB J 1994;8(2):217-225
Clinical Presentation	Device infections, pneumonia	UTI, septic shock, HAI	Influences infection source, severity, and treatment approach	Martin et al. N Engl J Med 2003;348(16):1546-1554

Proposed Bacterial Biomarkers panel

- Universal Bacterial Detection: Peptidoglycan fragments (N-acetylmuramic acid)

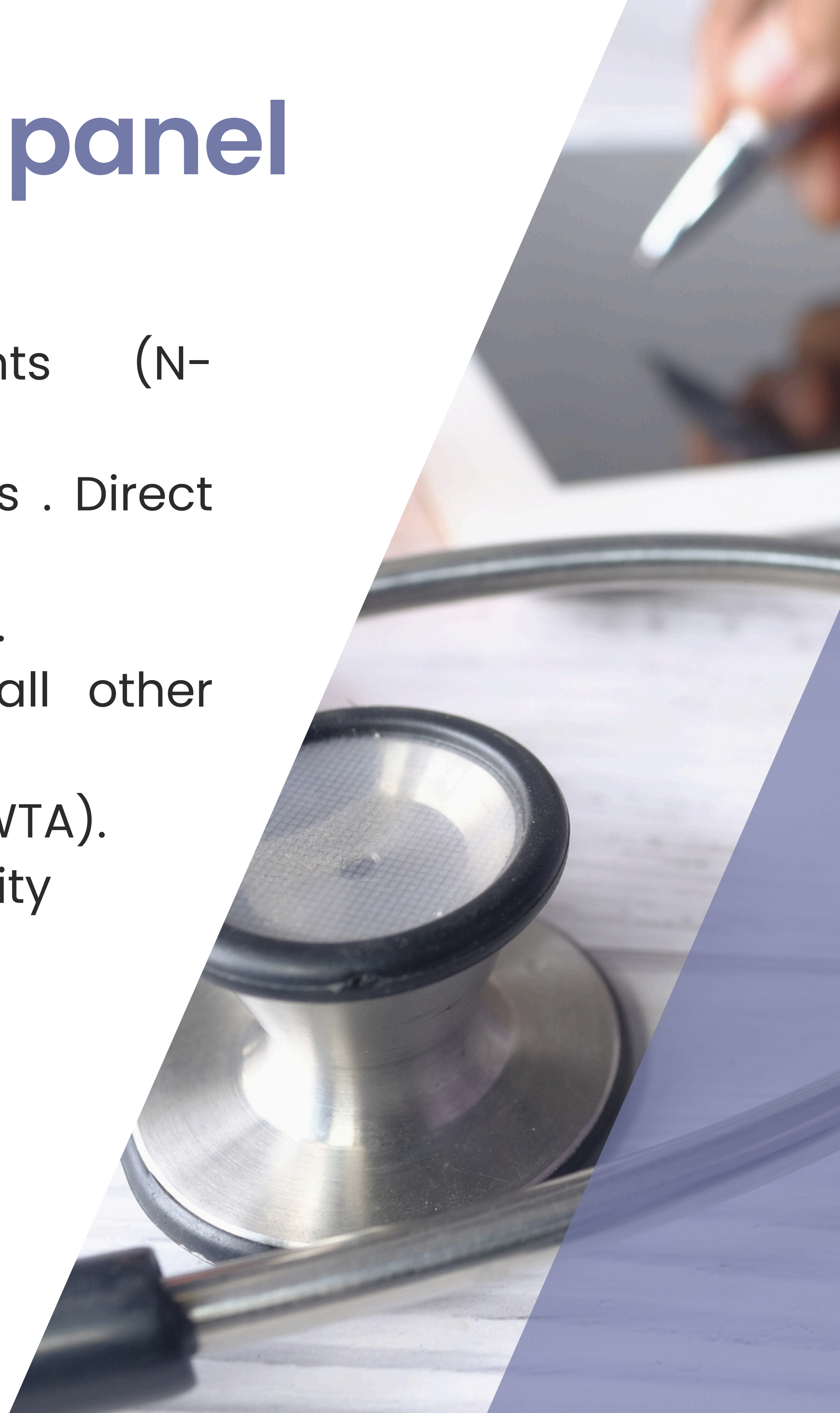
Reason : Present in ALL bacteria, absent in viruses/fungi/parasites . Direct bacterial confirmation.Unique to bacterial peptidoglycan only.

- Gram-negative: Lipopolysaccharide (LPS) — lipid A & O-antigen .

Reason :present Only in Gram-negative bacteria, absent in all other organisms.Strong enhancement, endotoxin correlation

- Gram-positive: Lipoteichoic acid (LTA) and wall teichoic acids (WTA).

Reason : present Only in Gram-positive bacteria , surface accessibility

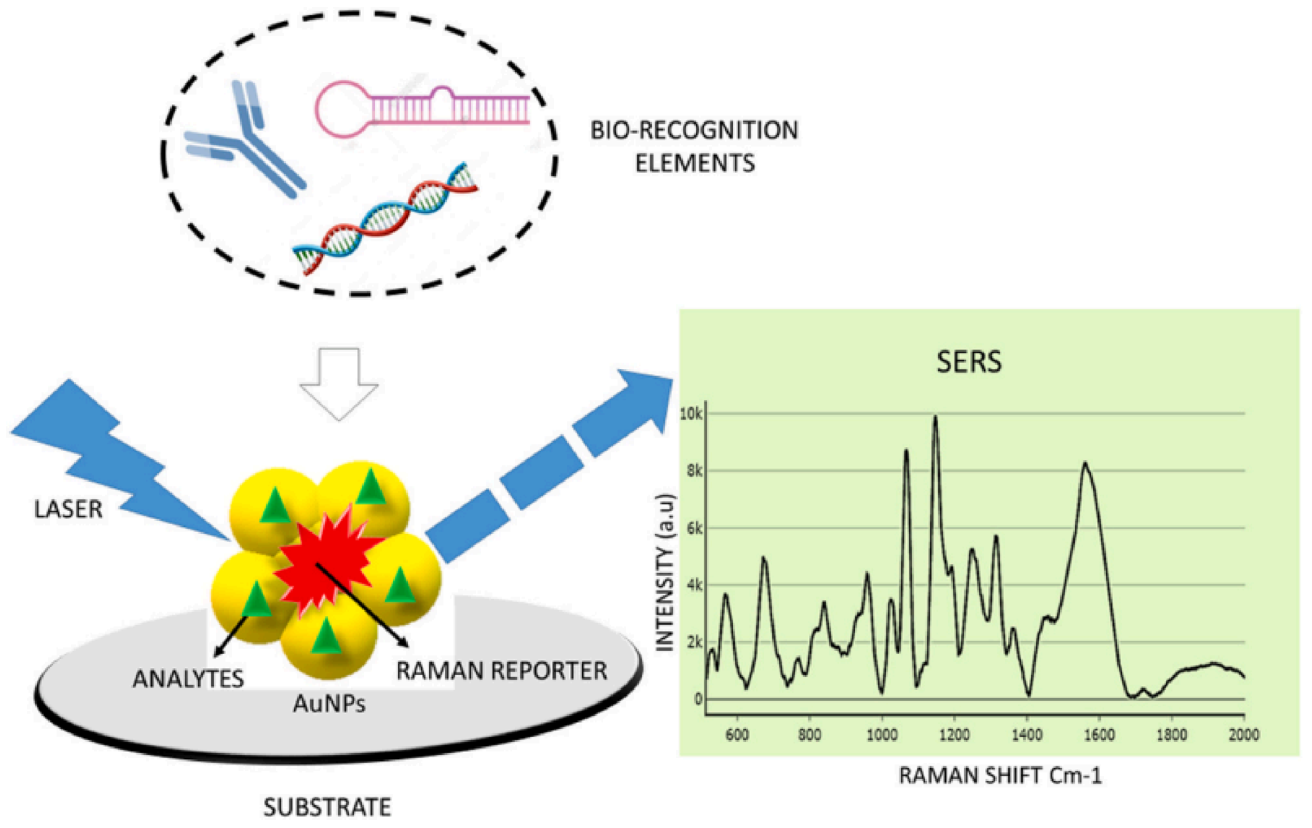


SERS Nanobiosensor: Overview

- SERS = Surface-Enhanced Raman Spectroscopy; provides molecular fingerprints.
- Plasmonic nanostructures (Ag/Au) create strong local fields (hotspots).
- Combine SERS substrate with capture probes (aptamers/antibodies) for selectivity.

SERS Fundamentals

- Raman scattering measures vibrational modes; specific peaks = molecule signature.
- Electromagnetic enhancement from plasmon resonance scales..
- Hotspots (nanogaps) give the largest enhancement; single-molecule sensitivity possible.



Nanomaterials & Substrate Design

• Choice of metal (Ag stronger SERS; Au more stable) affects resonance.
• Particle geometry (stars, rods, gaps) tunes hotspot intensity.
• Substrate uniformity & reproducible fabrication are essential for clinical use.

Excitation Wavelength & Practical Considerations

• Common lasers: 532 nm (resonant dyes), 633 nm, 785 nm (bio-friendly), 1064 nm (high autofluorescence samples).
• Balance: stronger scattering (shorter λ) vs. fluorescence/background and sample heating.
• Laser power and exposure time must be optimized for clinical samples.

Objective: To develop a reproducible plasmonic substrate with strong Raman enhancement for bacterial biomarker detection (LPS, LTA, PGN).

Nanosensor Substrate Fabrication Process

Substrate Selection

Choosing the base material for the substrate (glass, silicon, or polymer base)

Nanoparticle Synthesis

Gold and silver nanoparticles (AuNPs/AgNPs) via chemical reduction

Deposition Technique

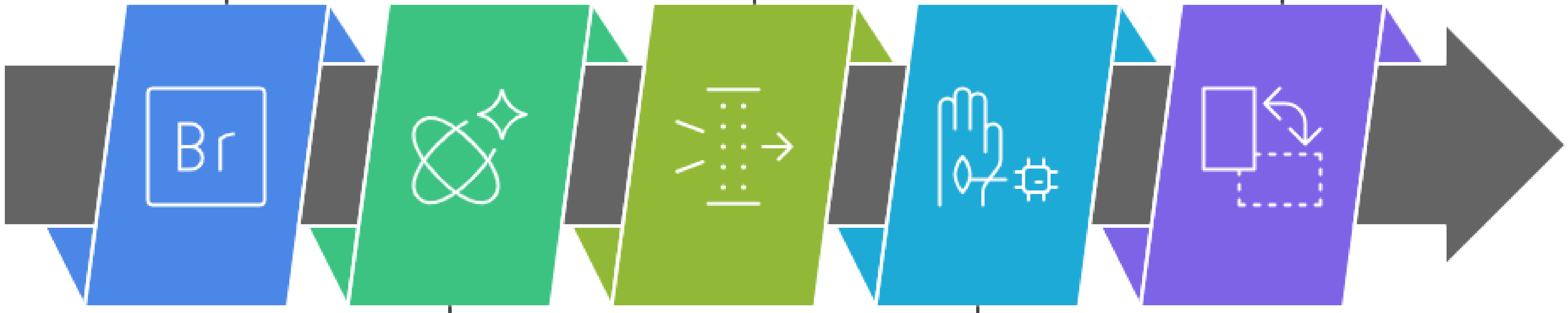
Drop-casting, spin-coating, or electrochemical deposition for uniform layer formation

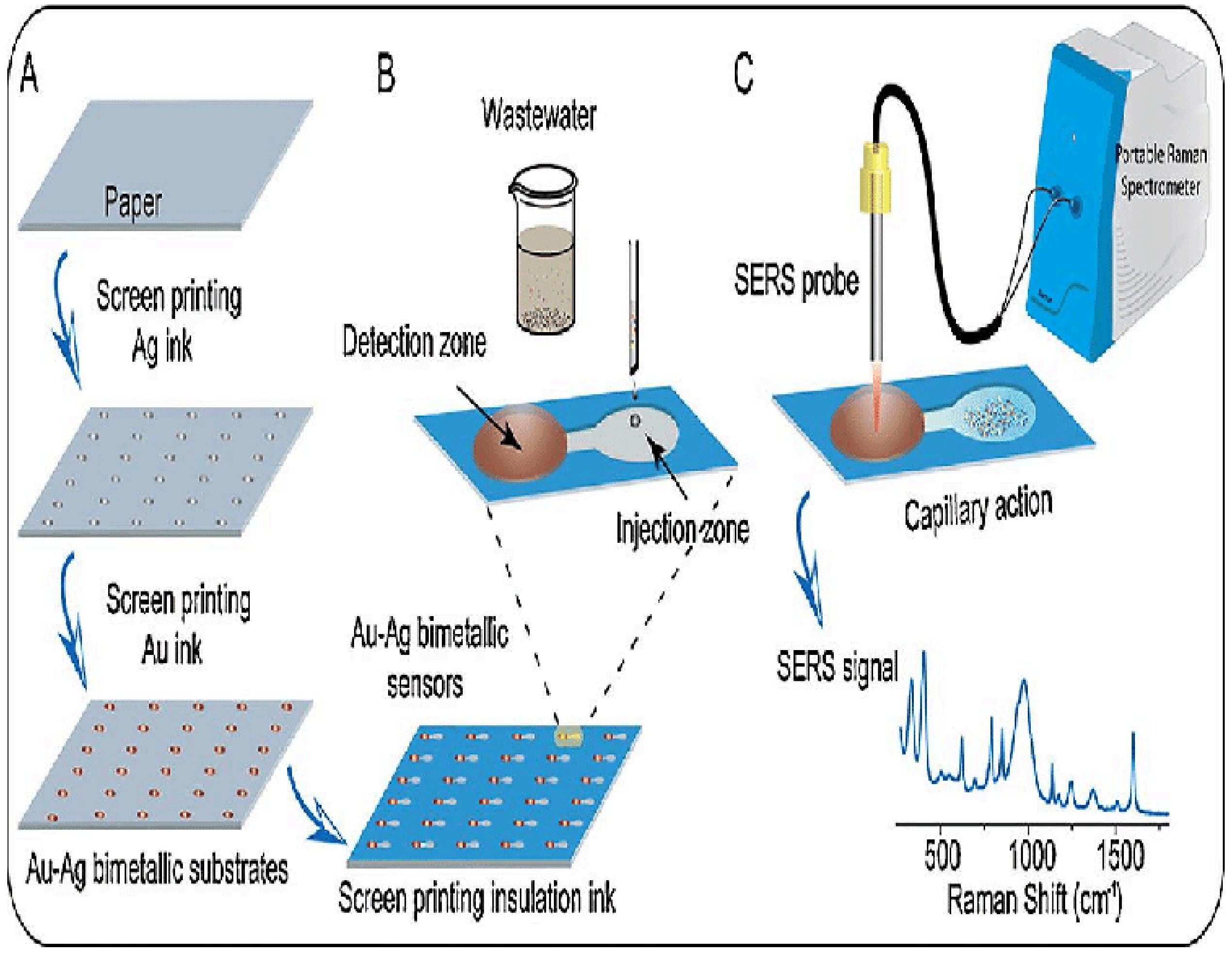
Surface Functionalization

Attaching biorecognition probes to the surface

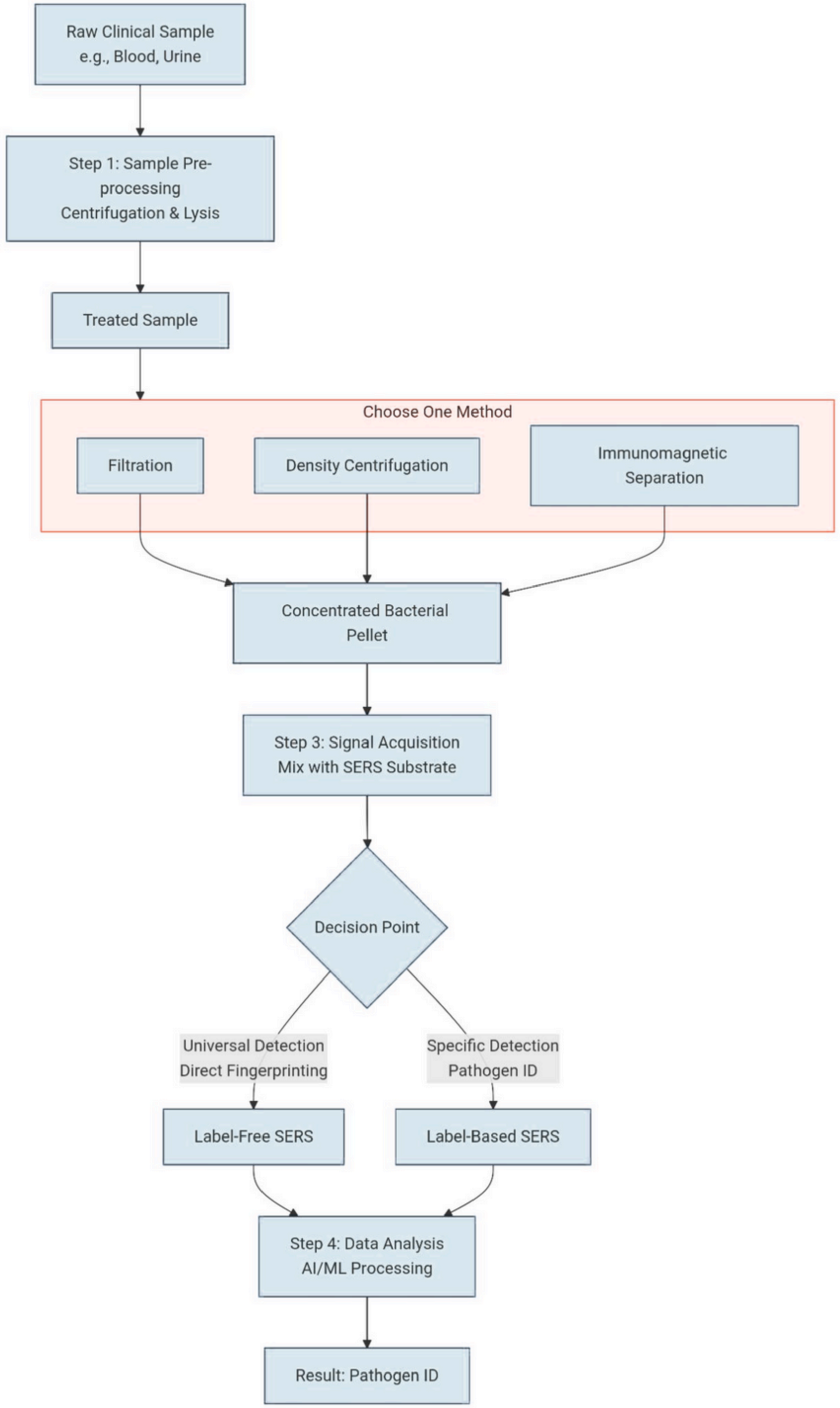
Outcome

Achieving a stable and enhanced substrate

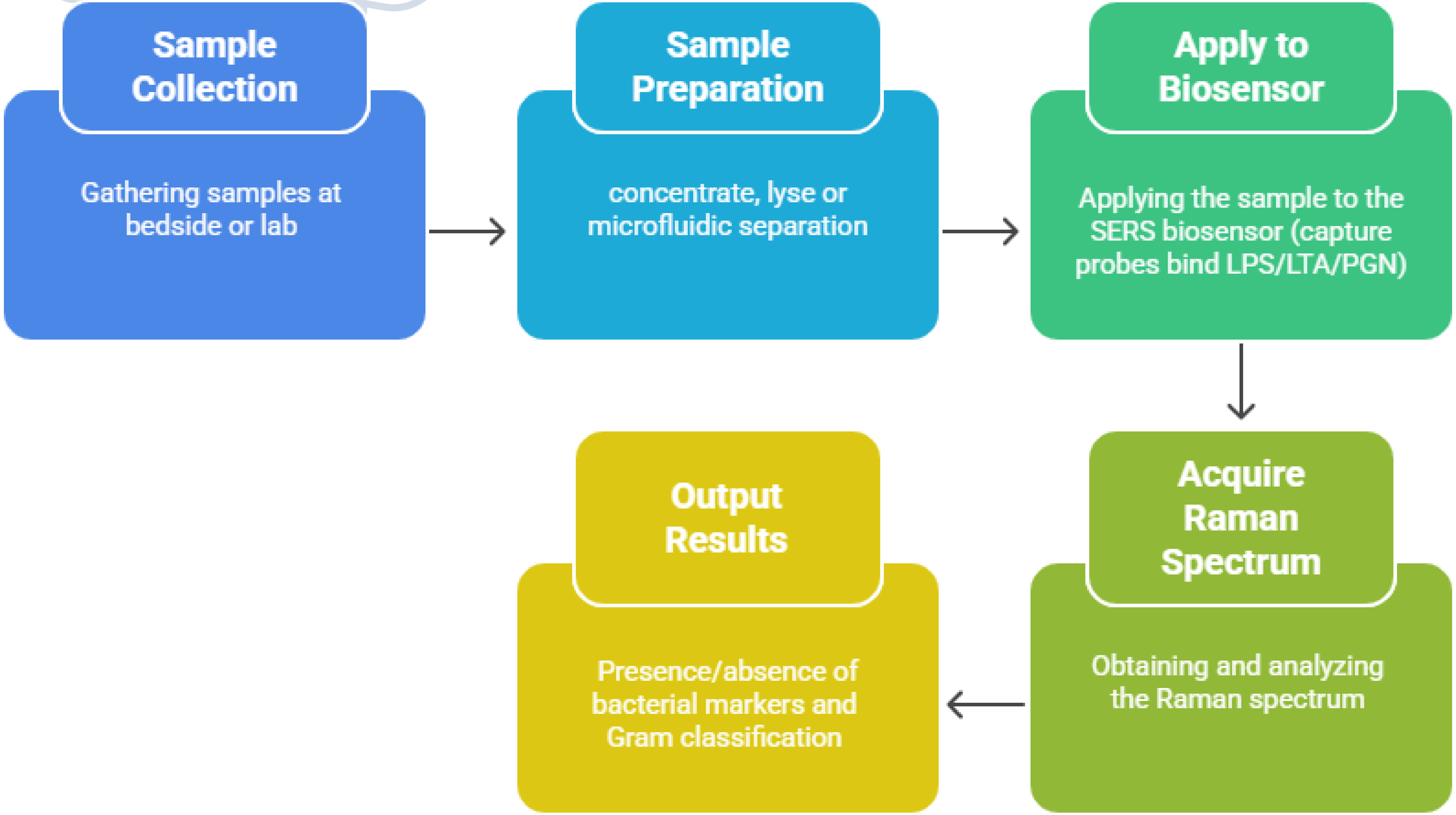




Workflow Pipeline:



Diagnostic Workflow (Patient → Answer)



Competitive Advantage & Market Differentiation :

Feature	Blood Culture	PCT/CRP Tests	Our SERS Biosensor
Time-to-Result	24-72 hours	~1 hour	< 30 MINUTES *
Pathogen Specificity	Yes	No	YES (Bacterial & Gram-Type)
AMR Guidance	Delayed	No	IMMEDIATE
Platform	Central Lab	Point-of-Care	POINT-OF-CARE

Key Message: With the help of this project we envision to bridge the gap between speed and specificity

Clinical Utility & Impact

• Rapid Gram-level info enables earlier targeted antibiotics (improves survival).
• Less empiric broad-spectrum use → supports antimicrobial stewardship.
• Portable SERS devices can enable near-patient, point-of-care testing in ED/ICU.

Challenges & Roadmap

- Extensive literature review done and finalization of biomarker process .
- SERS simulations initiation to optimize spectral response and substrate design.
- Next: experimental validation of simulation results using standard samples.
- Prototype fabrication and spectral benchmarking planned in next phase.
- Key challenges: ensuring signal reproducibility, substrate uniformity, and scaling for real-world use.

Long-term goal: translate optimized SERS workflow into a portable sepsis diagnostic device.

References

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Thank You